

# Advanced Warehouse Management In D365 Finance and Operations

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# Introduction

Microsoft Dynamics 365 Finance and Operations (D365 F&O) offer advanced warehouse management capabilities that aid businesses in improving their warehouse processes. This guide provides detailed steps for setting up and managing an advanced warehouse in D365 F&O.

This document is created with assumption that a warehouse has 5 inbound staging locations, 5 outbound staging location and rest of the warehouse area is floor used to store inventory in aisle, shelf, rack and bin. The actual inventory is not stored on License pallets.

## Step-by-Step Guide

### 1. Create a Storage Dimension Group

To create a storage dimension group:

- Navigate to Product information management > Setup > Dimension & variant groups > Storage dimension groups.
- Click New to create a new storage dimension group.
- Enter a Name and Description.
- Select the necessary storage dimensions such as Site, Warehouse, Location, and others as per image below.
- Save the storage dimension group.

Note: when you select use warehouse management processes to yes, system by default enables License Plate in setup.

The screenshot displays the 'Storage dimension groups' setup page in Microsoft Dynamics 365 Finance and Operations. The breadcrumb trail at the top indicates the path: Product information management > Setup > Dimension and variant groups > Storage dimension groups. The left-hand navigation pane lists several storage dimension groups, with 'SWL' (Site warehouse Location) selected. The main area shows the 'Storage dimensions' table, which lists various dimensions and their active status. Below this, the 'Warehouse specific setup' section includes a toggle for 'Use warehouse management processes' and a 'WAREHOUSE SPECIFIC SETUP' section with options for 'Mandatory' and 'Primary stocking'.

| Name             | Active | Blank recei... | Blank issue... | Physical in... | Financial i... | Coverage ... | For purcha... | For sales p... |
|------------------|--------|----------------|----------------|----------------|----------------|--------------|---------------|----------------|
| Site             | ✓      |                |                | ✓              | ✓              | ✓            | ✓             | ✓              |
| Warehouse        | ✓      |                |                | ✓              |                | ✓            | ✓             | ✓              |
| Location         | ✓      |                |                | ✓              |                |              |               |                |
| Inventory status | ✓      |                |                | ✓              |                |              |               |                |
| License plate    | ✓      | ✓              | ✓              | ✓              |                |              |               |                |

**Warehouse specific setup**

Use warehouse management pro... ☒ Yes

**WAREHOUSE SPECIFIC SETUP**

Mandatory ☐ No

Primary stocking ☒ Yes

## 2. Create a Tracking Dimension Group

To create a tracking dimension group:

- Navigate to Product information management > Setup > Dimension and variant groups > Tracking dimension groups.
- Click New to create a new tracking dimension group.
- Enter a Name and Description.
- Select the necessary tracking dimensions such as Batch number, Serial number, etc. as per image below.
- Save the tracking dimension group.

| Name         | Active                              | Active in sales process | Primary st... | Blank recei... | Blank issue... | Physical in...                      | Financial i...                      | Coverage ... | For purcha... | For sales p... | Transfer                            |
|--------------|-------------------------------------|-------------------------|---------------|----------------|----------------|-------------------------------------|-------------------------------------|--------------|---------------|----------------|-------------------------------------|
| Batch nu...  | <input checked="" type="checkbox"/> |                         |               |                |                | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> |              |               |                |                                     |
| Serial nu... |                                     |                         |               |                |                |                                     |                                     |              |               |                |                                     |
| Owner        |                                     |                         |               |                |                | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> |              |               |                | <input checked="" type="checkbox"/> |

## 3. Create a Warehouse and assign a name and site

To create a warehouse:

- Navigate to Warehouse management > Setup > Warehouse > Warehouses.
- Click New to create a new warehouse.
- Enter a Warehouse ID, Name, and assign a Site.
- Save the warehouse.

| Warehouse | Name       |
|-----------|------------|
| 80        | Burlington |

General

Site

2

Quarantine warehouse

28

Goods in transit warehouse

24-GIT

REFERENCE

Vendor account

Type

Default

Transit warehouse

27

Under delivery warehouse

24-UND

Master planning

Inventory and warehouse management

Default receipt location

Default production input location

Default shipment maintenance lo...

Work processing policy name

Default issue location

Issue status after raw material pi...

Picked

Default location for credit only re...

Default quality maintenance loca...

Default production finished goo...

Default kanban finished goods lo...

Default container type

Default location for final shipment

Default location for sorting

## 4. Assign Warehouse Management Attributes

To assign warehouse management attributes:

- Navigate to the Warehouse created in the earlier step.
- Under the General tab, select the necessary attributes such as Use warehouse management processes.
- Save the changes.

Standard view ▾

### Warehouses

Warehouse: 80 Name: Burlington

Hierarchy: Retail

Warehouse

Use warehouse management processes: ☒ Yes

Requirement for inventory reservation: Allow partial reservation ▾

REDUCE PICKED QUANTITY

Decrement load line: ☒ No

Enable quality order for warehouse: ☒ No

Print BOL before confirming shipment: ☒ No

Release rule failure option for audit: Warning and skip line ▾

INVENTORY

Default inventory status ID: ▾

Default container type: ▾

Work processing policy name: ▾

Auto update shipment: On quantity decrease ▾

RESERVATIONS

Reserve only warehouse in production: ☒ No

Reserve inventory at load posting: ☒ No

NEGATIVE INVENTORY

Financial negative inventory: ☒ No

Physical negative inventory: ☒ No

CYCLE COUNTING

Allow license plate moves during: ☒ No

INVENTORY STATUS CHANGE

Remove reservations and markings: None ▾

Remove inventory blocking: ☒ No

PRODUCTION ORDERS

Backflush must use reserved quantity: ☒ No

Posting method: Inherit ▾

HAZARDOUS GOODS REPORT LANGUAGE

Hazardous goods domestic report: ▾

Hazardous goods export report: ▾

## 5. Create Location Profiles

To create a location profile:

- Navigate to Warehouse management > Setup > Warehouse > Location Profiles.
- Click New to create a new location format.
- Enter a Location Profile ID and Name.
- Define the location format and location type and other attributes as per image below
- Save the location profile.
- Likewise create multiple location profiles as per requirement.

Finance and Operations Warehouse management > Setup > Warehouse > Location profiles USMF

Standard view ▾

### Location profiles

Location profile ID: InboundStage Name: Inbound Stage

General

Location format: STAGE ▾

Location type: STAGE ▾

Dock management profile ID: ▾

Use license plate tracking: ☒ No

Allow negative inventory: ☒ Yes

Allow mixed items: ☒ Yes

Enable location product dimensions: ☒ No

Allow mixed inventory statuses: ☒ No

Allow mixed inventory batches: ☒ Yes

Allow cycle counting: ☒ Yes

Generate check digit: ☒ No

Override rules for batch days: ☒ No

Allowed container type group: ▾

Enable item in location: ☒ No

Enable location activity date and time: ☒ No

Enable location status: ☒ No

Enable license plate positioning: ☒ No

Display mobile device LP position: ☒ No

## 6. Create Locations

To create locations formats and add segments:

- Navigate to Warehouse management > Setup > Warehouse > Locations.
- Click location setup wizard.
- Select warehouse, location profile id
- In lines define from number and To number.
- Click create.

Standard view

Location setup wizard

Warehouse

Location profile ID

Additional Zone 2

80

InboundStage

Zone ID

Additional Zone 1

Additional Zone 3

|  | Line number  | Segment description | From num... | To number | Static text | Length |
|--|--------------|---------------------|-------------|-----------|-------------|--------|
|  | 1.0000000000 | Stage               | 1           | 5         |             | 10     |

CreateCancel

Like wise create location for Outbound staging area as well as Storage floor.

## 7. Create Inbound Location Directives

To create inbound location directives:

- Navigate to Warehouse management > Setup > Location directives.
- Click New to create a new location directive.
- Select Inbound as the directive type.
- Define the criteria and actions for the location directive.
- Save the location directive.

Note: screenshot added in step for adding location directive.

## 8. Add Lines to the Location Directive

To add lines to the location directive:

- Navigate to the created location directive.
- Under the Lines tab, click Add line.
- Define the criteria for the line.
- Save the changes.

## 9. Add Location Directive Actions to the Location Directive

To add location directive actions:

- Navigate to the created location directive.
- Under the Actions tab, click Add action.
- Define the actions for the location directive.
- Save the changes.

The screenshot shows the 'Location directives' form in Microsoft Dynamics 365 Finance and Operations. The left pane shows a list of location directives: '1 Burlington Pick', '2 Burlington Put Single', and '3 Burlington Put Multiple'. The main pane shows the details for '1 Burlington Pick'. The 'Work type' is 'Pick', 'Directive code' is empty, and 'Scope' is 'Single item'. The 'Warehouse selection' section shows 'Warehouse' as '2' and 'Site' as '80'. The 'Lines' section shows a table with two rows:

| Sequence | From quantity | To quantity | Unit | Locate quantity | Restrict by unit | Round up to | Locate package | Allow split | Immediate repl |
|----------|---------------|-------------|------|-----------------|------------------|-------------|----------------|-------------|----------------|
| 1        | 0.00          | 10,000.00   | BALE | None            | ✓                |             |                |             |                |
| 2        | 0.00          | 10,000.00   | BALE | None            |                  |             |                | ✓           |                |

Note: In Pick no location directive needs to be added. In put location directive is needed. Update query as needed.

Similarly create Put single, put multiple and inbound location directives.

## 10. Create Work Template

To create a work template:

- Navigate to Warehouse management > Setup > Work > Work templates.
- Click New to create a new work template.
- Enter the necessary details for the work template.
- Save the work template.

Finance and Operations Warehouse management > Setup > Work > Work templates USMF

Work templates

Standard view \*~

Work order type  
Sales orders

Overview General

| Sequence | Work template         | Work template description | Valid                               | Automatic...             | Work pool ID | Work prior... |
|----------|-----------------------|---------------------------|-------------------------------------|--------------------------|--------------|---------------|
| 0        | Burlington SO Process | Ship direct               | <input checked="" type="checkbox"/> | <input type="checkbox"/> |              | 0             |

Work Template Details

+ New Delete Move up Move down Work line breaks

| Line number  | Work type | Custom work type | Mandatory                           | Stop work                | Freeze                   | Directive code | Work class ID | Quality check temp... | Anchoring gro |
|--------------|-----------|------------------|-------------------------------------|--------------------------|--------------------------|----------------|---------------|-----------------------|---------------|
| 1.0000000000 | Pick      |                  | <input checked="" type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |                | Sales         |                       |               |
| 2.0000000000 | Put       |                  | <input checked="" type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | ToOutLoc       | Sales         |                       |               |

## 11. Create Wave Template

To create a wave template:

- Navigate to Warehouse management > Setup > Waves > Wave templates.
- Click New to create a new wave template.
- Enter the necessary details for the wave template.
- Save the wave template.

Finance and Operations Warehouse management > Setup > Waves > Wave templates USMF

Wave template

Wave template name Wave template description Wave template sequence Template status

Burlington Shipping Burlington Shipping 1 Valid

Warehouse selection Warehouse 80

Warehouse selection Site Warehouse

Warehouse 2 80

General

**WAVE TEMPLATE INFORMATION**

Automate wave creation ☒ Yes

Assign to open waves ☐ No

Process wave at release to wareh... ☒ Yes

Process wave automatically at th... ☐ No

Automate wave release ☒ Yes

Automate replenishment work re... ☒ Yes

Continue wave processing when ... ☐ No

**WAVE THRESHOLDS**

Wave weight threshold 0.00

Shipment threshold 0

Line threshold 0

**DEFAULT VALUES**

Attribute 1

Attribute 2

**WAVE EXECUTION NOTIFICATIONS**

Wave notification policy

Methods

## 12. Configure warehouse management parameters

To configure warehouse management parameters:

- Navigate to Warehouse management > Setup > warehouse management parameters
- Update location types as per requirement



Finance and Operations | Warehouse management > Setup > Warehouse management parameters | USMF

Standard view

### Warehouse management parameters

**General**

Set up parameters for Warehouse management

**Company information**

**Location profiles**

User Location Profile: USER

**Location types**

Staging location type: STAGE | Packing location type: FLOOR | Final shipping location type: STAGE | Sorting location type:

**Receiving**

Exception for missing items from...: ASN | Transfer order receiving process: Combine the registration a... | Enable license plate receiving his...: No

- Configure wave parameters as per requirement

Finance and Operations | Warehouse management > Setup > Warehouse management parameters | USMF

Standard view

### Warehouse management parameters

**General**

Set up parameters for Warehouse management

**Wave processing**

Enable schedule work creation: ☒

Wave processing batch group:

Process waves in batch: ☒

Create wave progress log: ☒ Yes

Create wave processing history log: ☒ Yes

Create containerization history log: ☒ Yes

Wait for lock (ms): 60000

Disable wave processing batch n...: ☒ No

Put location concurrency strictness: Performance optimized

Wait for put location directive lo...: 0

Ensure other parameters are also set correctly,

## 13. Update default locations on warehouse

To update default location for receipt and final location,

- Navigate to Warehouse management > Setup > Warehouse > Warehouses.
- Filter required warehouse
- Navigate to inventory and warehouse management tab and update location

## 14. Create a Catch Weight Handling Policy

To create a catch weight handling policy:

- Navigate to Product information management > Setup > Catch weight policies.
- Click New to create a new catch weight handling policy.
- Enter a Name and Description.
- Define the settings for the policy.
- Save the catch weight handling policy.

## 15. Create a New Released Product with catchweight attributes

To create a new released product:

- Navigate to Product information management > Products > Released products.
- Click New to create a new released product.
- Enter the necessary product details.
- Assign the right storage and tracking dimension groups.
- Save the released product.

## 16. Create Purchase Order and Receive It

To create a purchase order:

- Navigate to Procurement and sourcing > Purchase orders > All purchase orders.
- Click New to create a new purchase order.
- Enter the vendor and product details.
- Confirm the purchase order and receive the items.

## 17. Create Sales Order and Process It via Advanced Warehouse Management

To create a sales order:

- Navigate to Sales and marketing > Sales orders > All sales orders.
- Click New to create a new sales order.
- Enter the customer and product details.
- Confirm the sales order and process it using the advanced warehouse management features.

## Conclusion

Setting up and managing an advanced warehouse in D365 Finance and Operations involves several detailed steps that ensure the warehouse functions efficiently. By following this guide, you can streamline your warehouse processes and improve overall productivity.

### Important Notes:

By design advance warehouse functionality is enabled, in storage dimensions license plates are enabled by default. Using this guide we can store inventory without license plates (LP).

However, system requires LP for processing transaction. Hence to successfully process transactions, we have designated one location on inbound staging area and one location at outbound staging area as pallet enabled location. In order to consumes pallet at put locations, we have automated pallet creation in warehouse device using below showcased out of the box functionalities.

| Warehouse | Location   | Check digits | Location profile ID | Zone ID | Additional Zone 1 | Additional Zone 2 | Additional Zone 3 | Work policy | Last cycle |
|-----------|------------|--------------|---------------------|---------|-------------------|-------------------|-------------------|-------------|------------|
| 80        | 0000000001 | STAGE        |                     |         |                   |                   |                   |             |            |

Standard view

### Mobile device menu items

Menu item name: Sales Picking Title: Sales Picking Mode: Work Use existing work: ☒ Yes

**General**

|                                                                   |                                                                |                                                              |                                                          |
|-------------------------------------------------------------------|----------------------------------------------------------------|--------------------------------------------------------------|----------------------------------------------------------|
| Display inventory status: <input checked="" type="checkbox"/> Yes | Anchoring: <input type="checkbox"/> No                         | Count total item quantity first: <input type="checkbox"/> No | Allow over pick: <input checked="" type="checkbox"/> Yes |
| Directed by: User directed                                        | Override target license plate: <input type="checkbox"/> No     | Disable cycle count thresholds: <input type="checkbox"/> No  | Group pick: <input type="checkbox"/> No                  |
| Use process guide: <input type="checkbox"/> No                    | Override license plate during put: <input type="checkbox"/> No | Group put away: <input type="checkbox"/> No                  | Deferred put processing policy: Respect settings         |
| Barcode data policy:                                              | Allow splitting of work: <input type="checkbox"/> No           | Pick oldest batch: None                                      | Fast validation: <input type="checkbox"/> No             |
| Pick and pack: <input type="checkbox"/> No                        | Keep work locked by user: <input type="checkbox"/> No          | Audit template ID:                                           | Show work line list: Never show                          |
| Generate license plate: <input checked="" type="checkbox"/> Yes   | Display summary of pick screen: <input type="checkbox"/> No    | Handle by license plate: <input type="checkbox"/> No         |                                                          |

**Work classes**

+ New Delete

Finance and Operations Warehouse management > Setup > Mobile device > Mobile device menu items USMF

Standard view

### Mobile device menu items

Menu item name: Purchase Receive Title: Purchase Receive Mode: Work Use existing work: ☐ No

**General**

|                                                      |                                                                 |                                                                  |                                                                    |
|------------------------------------------------------|-----------------------------------------------------------------|------------------------------------------------------------------|--------------------------------------------------------------------|
| Use default data: <input type="checkbox"/> No        | License plate grouping policy: None                             | Display batch disposition code: <input type="checkbox"/> No      | Generate catch weight tag: <input checked="" type="checkbox"/> Yes |
| Work creation process: Purchase order line receiving | Generate license plate: <input checked="" type="checkbox"/> Yes | Print label: <input type="checkbox"/> No                         | Assign putaway cluster: <input type="checkbox"/> No                |
| Use process guide: <input type="checkbox"/> No       | Work template:                                                  | Block receiving with ASN item in...: <input type="checkbox"/> No | Deferred receiving policy ID:                                      |
| Barcode data policy:                                 | Display disposition code: <input type="checkbox"/> No           | Display container type: <input type="checkbox"/> No              |                                                                    |

## Inventory reservation:

System is configured to consume inventory on FIFO basis, if the inventory on location is partially available, system will automatically consume inventory from next available location and create pick work transaction fro same.